

Yu Zhu

AI for Science · Computational Chemistry · Computer Vision

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1 Research Profile

Research at the intersection of AI for Science, computational chemistry, and computer vision, spanning machine learning force fields, physics-driven reconstruction of molecular orbitals from scanning tunneling microscopy images, open-set fine-grained recognition, and materials modeling.

2 Education

M.Sc., Department of Chemistry
Fudan University

Sep 2023 – Jun 2026
Shanghai, China

- Supervisors: Prof. Xin Xu and Young Researcher Sai Duan.
- Research: machine learning force fields, open-set recognition, and scanning tunneling microscopy.

B.Sc., College of Chemistry, Chemical Engineering and Materials Science
Soochow University

Sep 2019 – Jun 2023
Suzhou, China

- GPA: 3.8/4.0 (92.1/100).
- Supervisors: Prof. Jianlin Yao and Assoc. Prof. Minmin Xu.

3 Experience

AI for Science Intern
Microsoft Research

Jul 2024 – Jun 2025
Shanghai, China

- Contributed benchmarks and scalable computational workflows for materials modeling.
- Evaluated universal machine learning force fields and ran inference on models up to 1B parameters.
- Managed high-throughput datasets with up to 200M entries and built Azure pipelines supporting up to 10K concurrent tasks.

4 Selected Projects

STM-Net: Reconstructing Molecular Orbitals from STM Images

Jan 2024 – Jul 2024

- Simulated STM images at the DFT level using Bardeen's approximation.
- Developed a physics-driven U-Net to separate s-wave molecular-orbital signals from p-wave CO-tip contributions.
- Reached PSNR above 37 dB using 10% of the training data and above 43 dB on the full dataset.

MatterSim: Materials Modeling Workflows

Jul 2024 – Jun 2025

- Built benchmarks and high-throughput workflows using VASP, atomate2, phonopy, phono3py, ESEN, MACE, and ORB.

MPBR for Open-Set Fine-Grained Recognition

Jan 2025 – Jun 2025

- Developed benchmark pipelines and evaluated OSR models on iNat2021-OSR using t-SNE, Grad-CAM, Swin Transformer, and HAN-OSFGR.

SERS Imaging for Fingerprint Identification

Apr 2021 – Jun 2023

- Performed SERS imaging and chemical mapping of latent and live fingerprints on flexible substrates.

5 Publications

- Yu Zhu**, R. Xue, H. Ren, Y. Chen, W. Yan, B. Wu, S. Duan, H. Zhang, L. Chi, and X. Xu. [Reconstructing Pristine Molecular Orbitals from Scanning Tunneling Microscope Images via Artificial Intelligence Approaches](#). *JACS Au*, 5(7), 3163–3170, 2025.
- J. Li, Z. Chen, Q. Wang, H. Yang, Z. Lu, G. Li, S. Chen, **Yu Zhu**, et al. [Probing the Limit of Heat Transfer in Inorganic Crystals with Deep Learning](#). arXiv:2503.11568, 2025.
- J. Tan, P. Jing, **Yu Zhu**, and Y. Liu. [MPBR: Multimodal Progressive Bidirectional Reasoning for Open-Set Fine-Grained Recognition](#). *ICCV*, 1282–1291, 2025.
- M. Wu, J. Tan, **Yu Zhu**, and J. Ren. [KappaFormer: Physics-aware Transformer for Lattice Thermal Conductivity via Cross-domain Transfer Learning](#). arXiv:2604.03547, 2026.

6 Honors & Awards

- China National Scholarship, top 0.2% (2025 and 2022).
- Silver Award, China International College Students' Innovation and Entrepreneurship Competition, top 0.02% (2021).
- First Prize, Chinese Mathematics Competition for College Students, top 5.3% (2020).

7 Technical Skills

Scientific computing: Python, VASP, atomate2, joblib, phonopy, phono3py.

Machine learning: U-Net, DAE, Mask2Former, ESEN, MACE, ORB, Swin Transformer, HAN-OSFGR.

Scientific methods: STM simulation, Bardeen approximation, Tersoff-Hamann theory, Chen derivative rule, SERS mapping.

Visualization: t-SNE, Grad-CAM, Origin.